

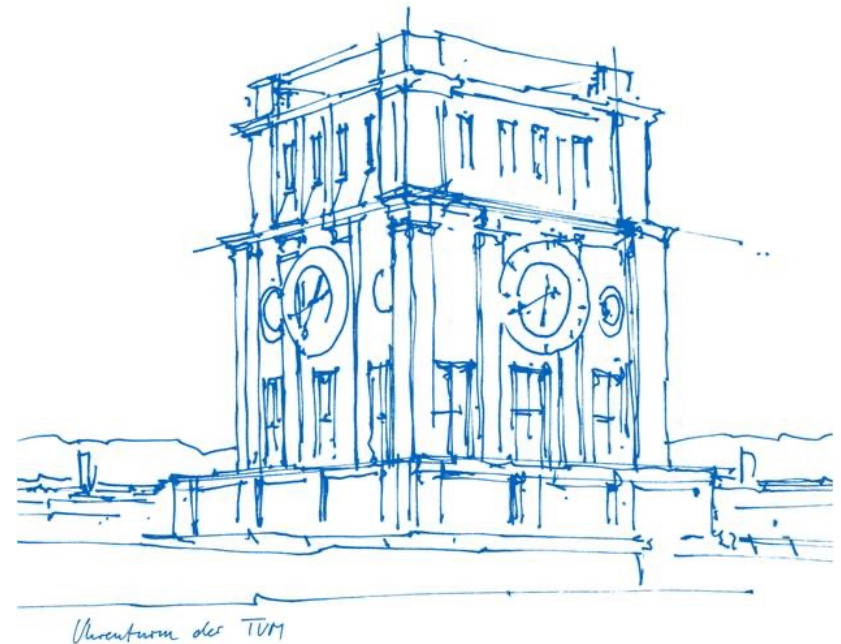
Finding Stable Price Zones in European Electricity Markets: Aiming to Square the Circle?

Teodora Dobos, Martin Bichler, Johannes Knörr

TUM School of Computation, Information and Technology

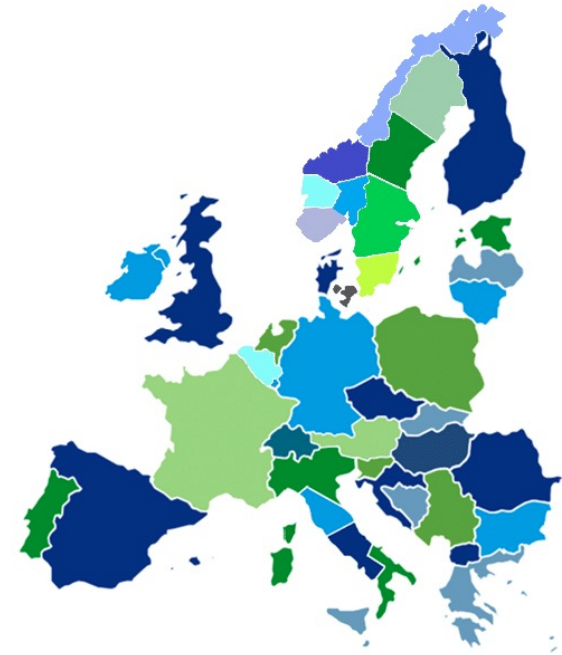
International Ruhr Energy Conference

August 28, 2024



European electricity market

- **Zonal** market design
- Germany forms **a large bidding zone**
 - **uniform price** in the day-ahead market



Source: <https://fsr.eui.eu/electricity-markets-in-the-eu/>

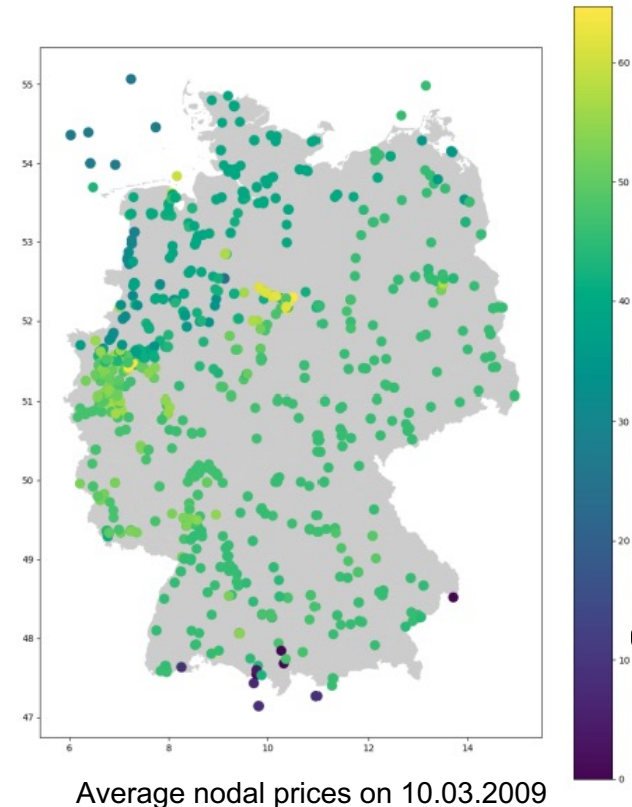
Zonal market design: negative implications



Poor local incentives



Welfare losses



Zonal market design: negative implications



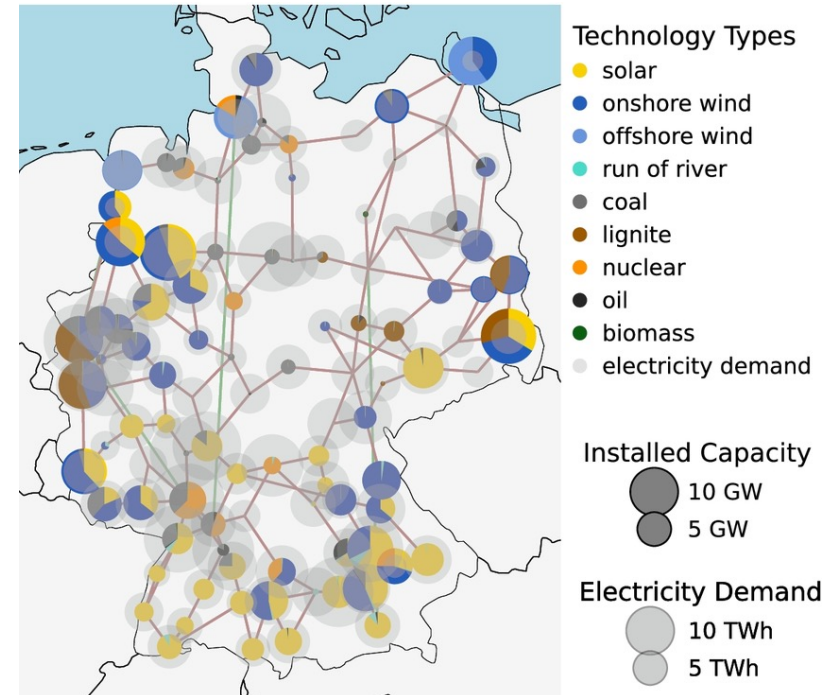
Poor local incentives



Welfare losses



Congestion & large redispatch costs



Source: *Net load forecasting using different aggregation levels*, 2022, Beichter et. al.

Zonal market design: negative implications



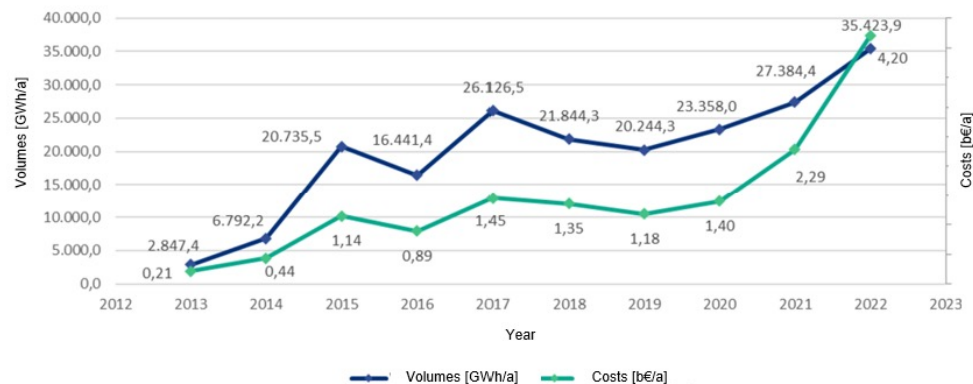
Poor local incentives



Welfare losses



Congestion & large redispatch costs

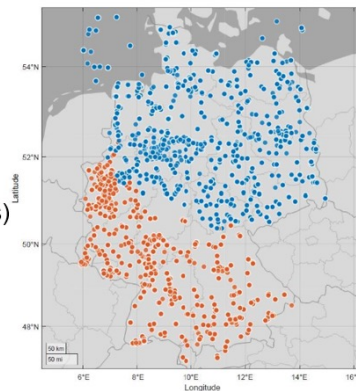


Source: : BDEW 2022, Bundesnetzagentur 2023c

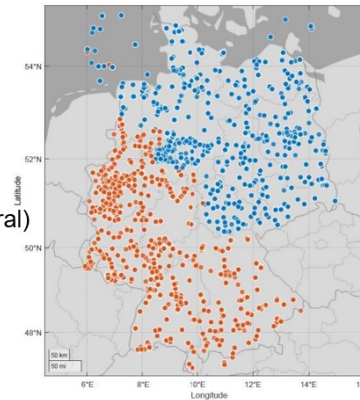
Potential solutions

- ACER: **4 alternative bidding zone configurations** (*Bidding Zone Review**, 2019)
 - based on the results of the *Locational Marginal Pricing Study* (2022)

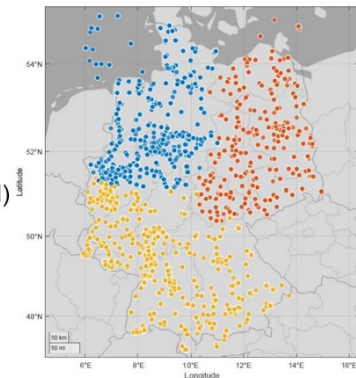
DE2 (k-means)



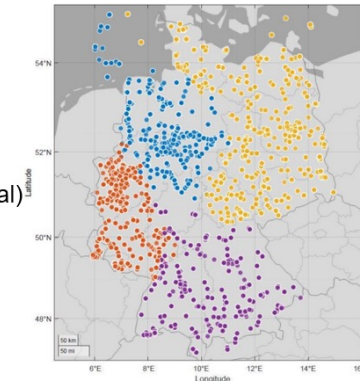
DE2 (spectral)



DE3 (spectral)



DE4 (spectral)



*Final results expected at the end of the year.

Configurations objectives

1. Ensure efficient **congestion management** & **market efficiency**
2. Should be **based on long-term, structural congestions** in the network
3. Should **minimize the price dispersion** within each zone

How *beneficial* are the proposed configurations?

How *stable* are the proposed configurations?

*Objectives acc. to Commission Regulation (EU) 2015/1222 and 2019/943

Clustering problem (ACER)

Data points: nodes (substations)

Features: time series of nodal prices:

- 3 years x 8 weeks x 7 days x 12 hours = 2,016 prices per node

Algorithms:

1. Constrained K-Means
2. Spectral Clustering with constrained K-Means
3. Spectral Clustering with constrained deterministic iterative refinement clustering

Our clustering problem

Data points: nodes (substations)

Features: time series of nodal prices **in a specific year**

- 8 weeks x 7 days x 12 hours = 672 prices per node
- **Latitude and longitude**

Algorithms:

1. Constrained K-Means
2. Spectral Clustering with constrained K-Means

Our clustering problem

Data points: nodes (substations)

Features: time series of nodal prices **in a specific year**

- 8 weeks x 7 days x 12 hours = 672 prices per node
- **Latitude and longitude**

Algorithms:

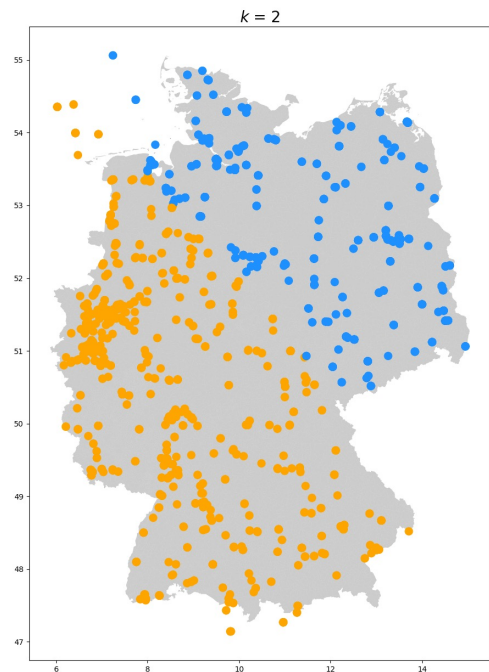
1. Constrained K-Means
2. Spectral Clustering with constrained K-Means

Can we reproduce the configurations proposed by ACER?

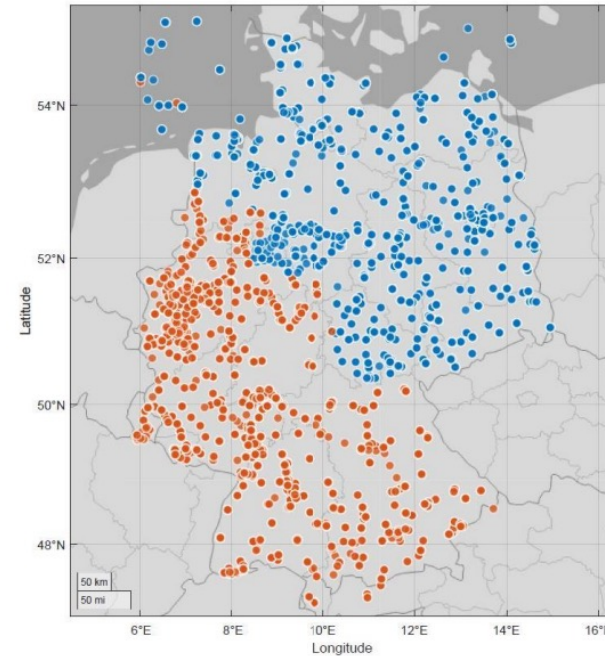
Exercise: cluster based on **latitude & longitude**

Clustering based on latitude & longitude

K-Means based on latitude & longitude

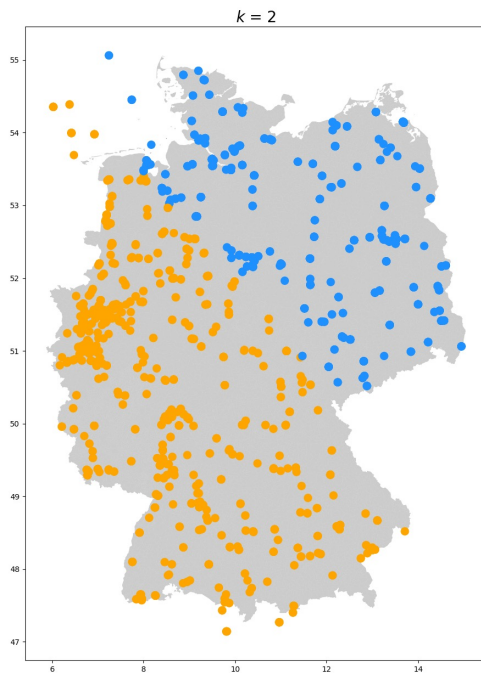


DE2 (spectral)

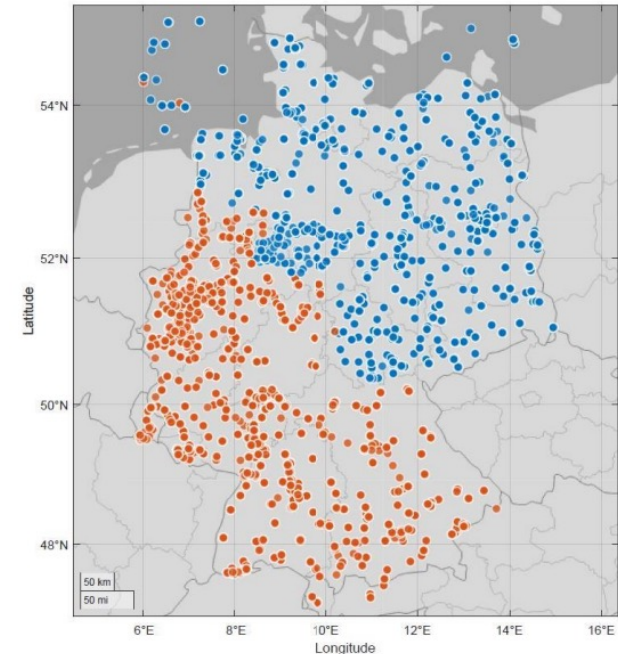


Clustering based on latitude & longitude

K-Means based on latitude & longitude



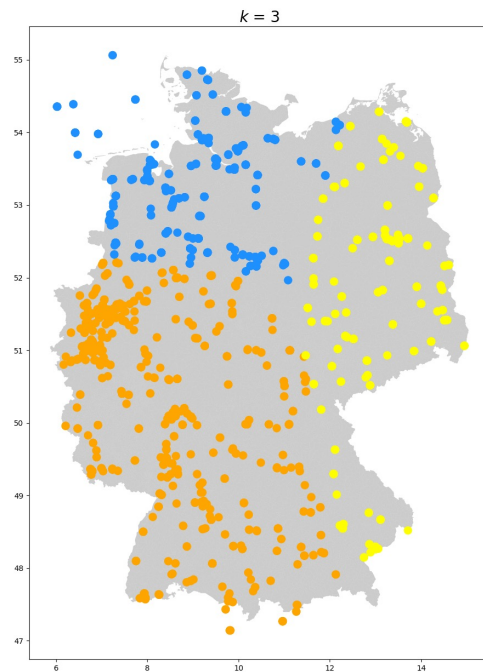
DE2 (spectral)



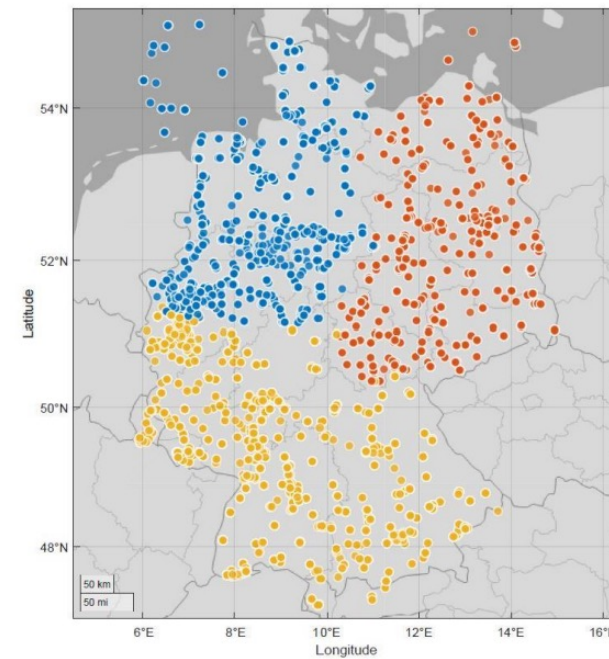
Source: <https://www.germanguide4u.com/federal-states/>

Clustering based on latitude & longitude

K-Means based on latitude & longitude

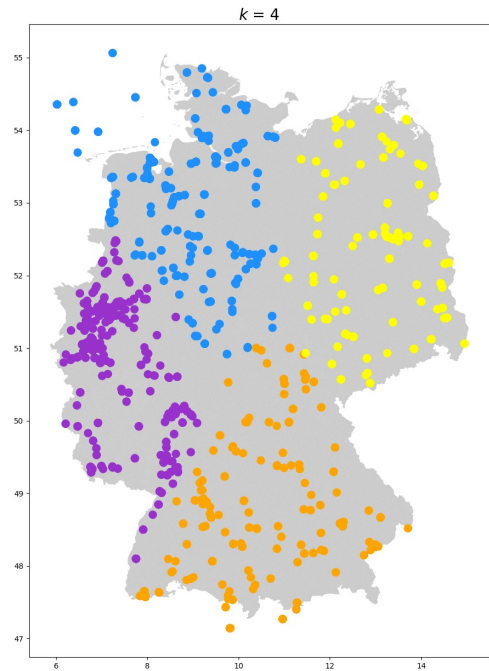


DE3 (spectral)

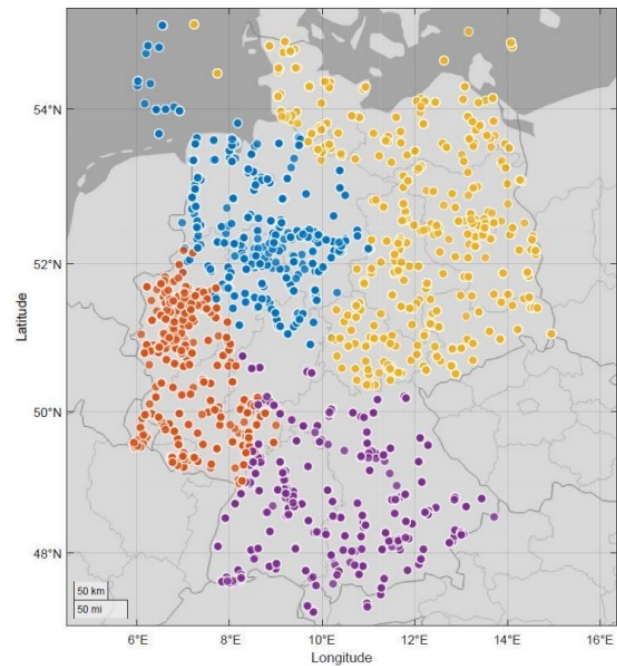


Clustering based on latitude & longitude

K-Means based on latitude & longitude



DE4 (spectral)



What about the **prices**?

Next: Stability evaluation w.r.t. prices

3 criteria

Stability evaluation (I)

1. **Intra- and inter-cluster similarity** - For a clustering $\mathcal{C} = \{C_1, \dots, C_k\}$ consider:

- Average distance between cluster mean prices

$$d_{\mu}(C_i, C_j) = d(\mu_i, \mu_j)$$

price absolute difference

- Nearest neighbor cluster distance

$$d_{\min}(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y)$$

- Price standard deviation $\sigma(C_i)$ inside a cluster C_i

Stability evaluation (II)

1. Intra- and inter-cluster similarity
2. **Temporal stability**

For two clusterings $\mathcal{C}_1$ and $\mathcal{C}_2$ computed for two periods t_1 and t_2 , respectively, consider the

Rand Index:

$$RI(\mathcal{C}_1, \mathcal{C}_2) = \frac{\sum_{1 \leq l < r \leq N} \gamma(x_l, x_r)}{\binom{N}{2}} \in [0, 1]$$

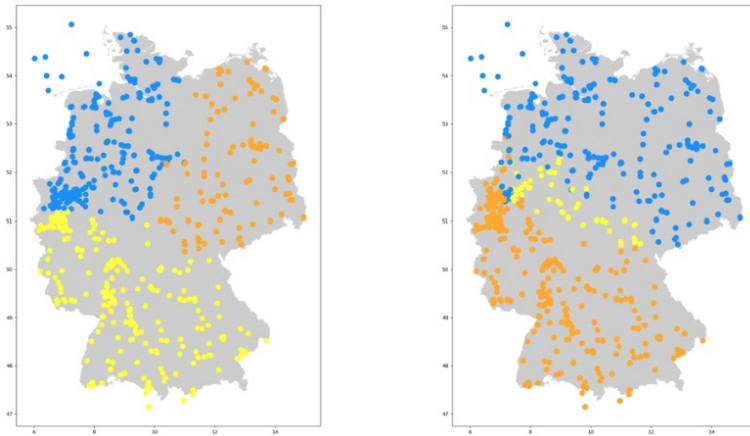
where

$$\gamma(x_l, x_r) = \begin{cases} 1 & \text{if nodes } x_l, x_r \text{ in the **same** cluster in } \mathcal{C}_1, \mathcal{C}_2 \\ 1 & \text{if nodes } x_l, x_r \text{ in **different** clusters in } \mathcal{C}_1, \mathcal{C}_2 \\ 0 & \text{otherwise.} \end{cases}$$

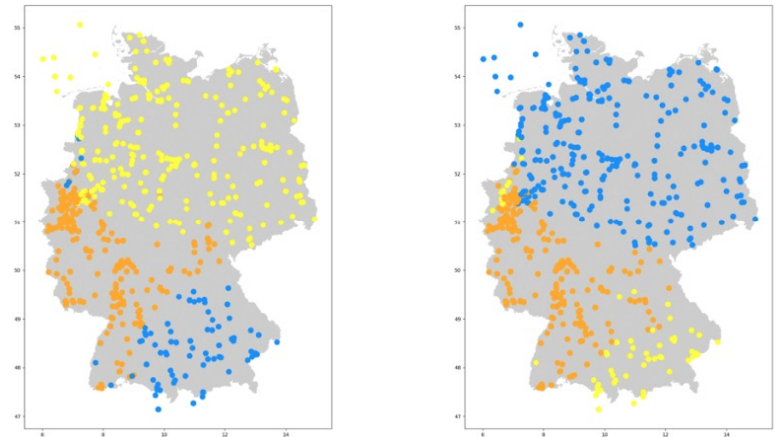
Stability evaluation (II)

1. Intra- and inter-cluster similarity
2. **Temporal stability**

$RI = 0.44$



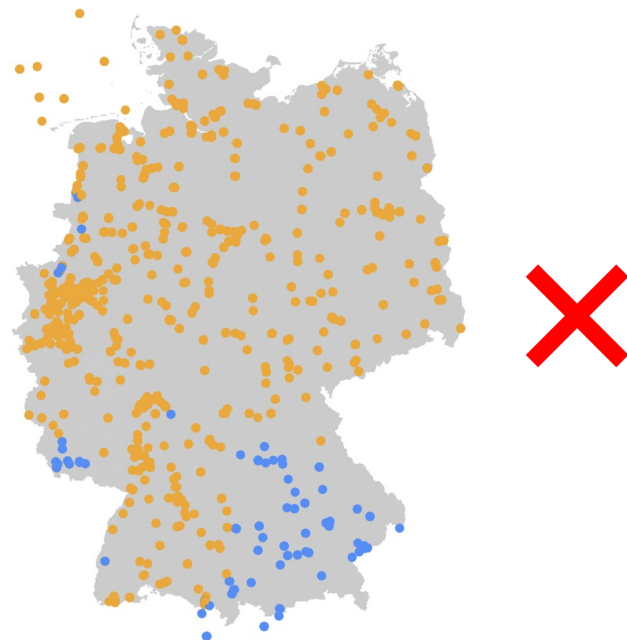
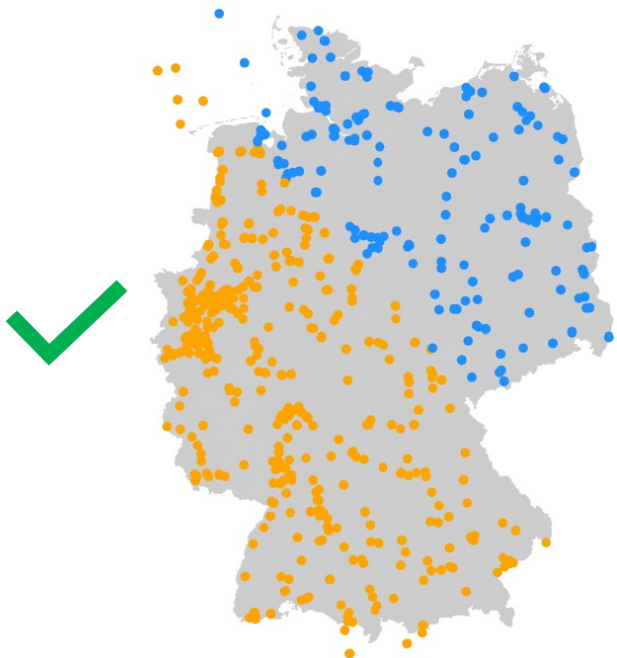
$RI = 0.93$



Stability evaluation (III)

1. Intra- and inter-cluster similarity
2. Temporal stability
3. **Spatial coherence**

1. zones are “linearly separated” on the map

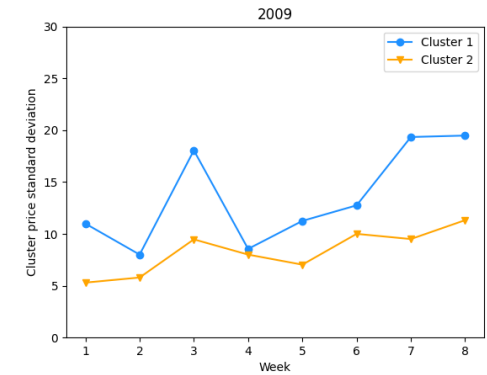
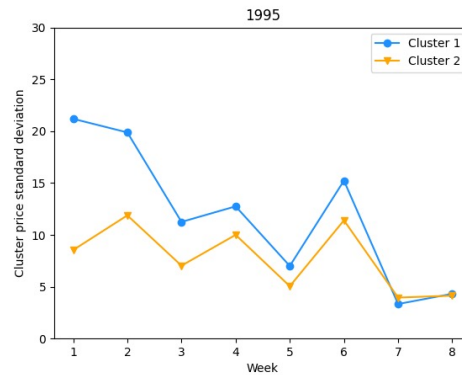
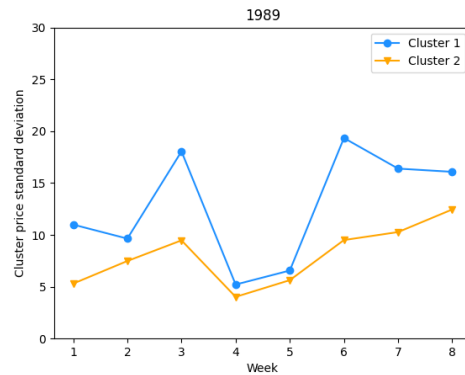
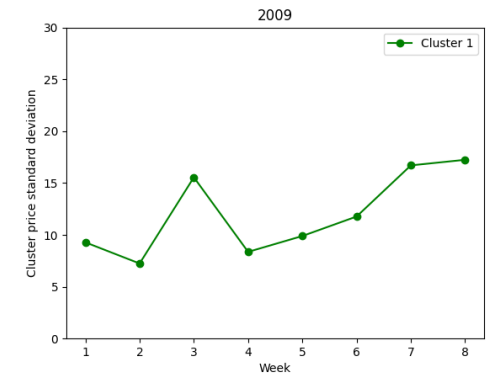
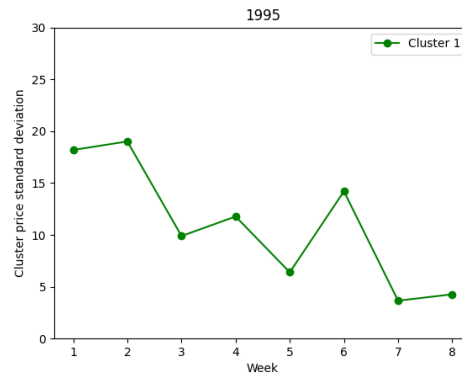
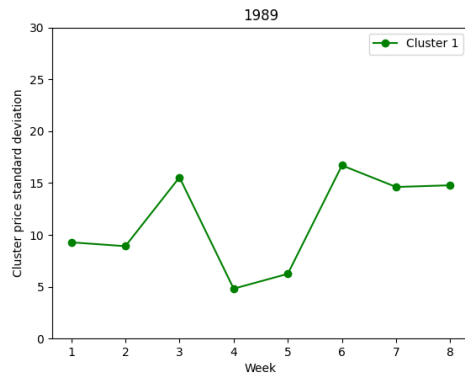


Stability evaluation (III)

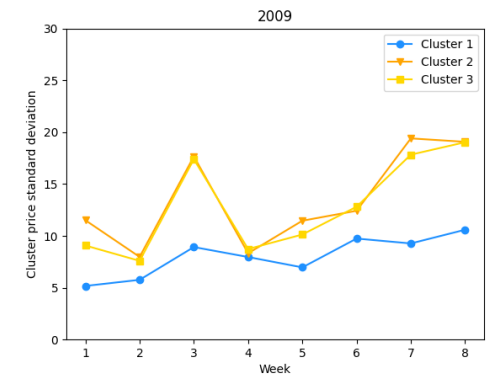
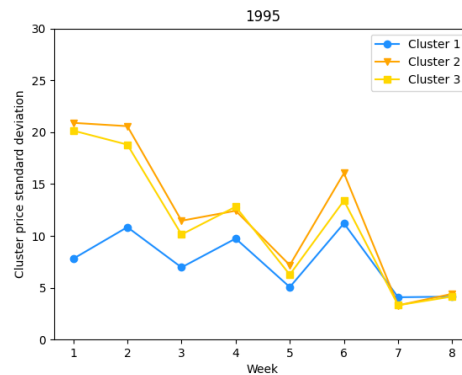
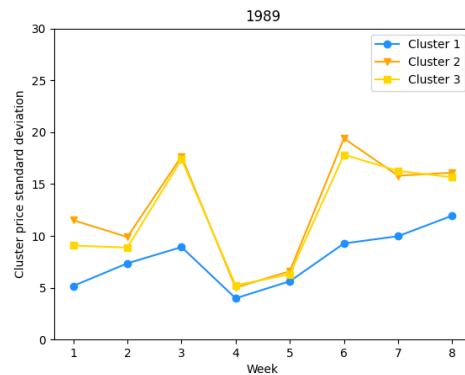
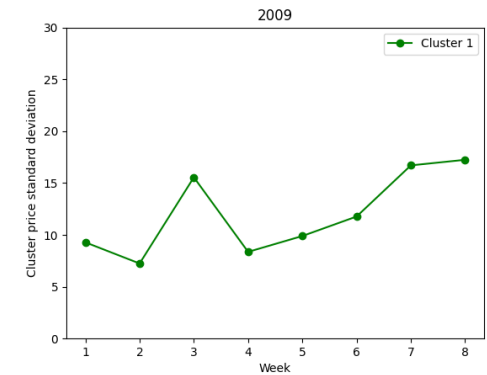
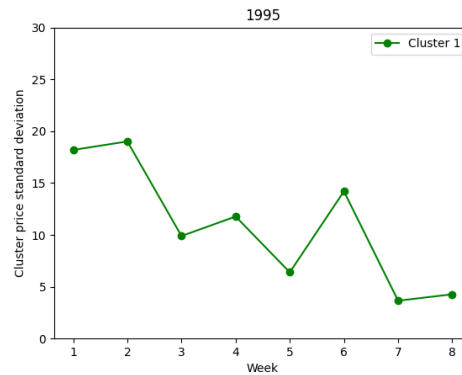
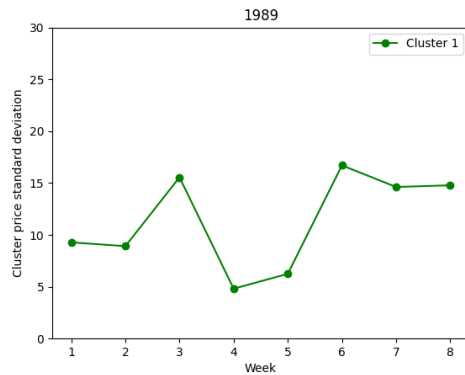
1. Intra- and inter-cluster similarity
2. Temporal stability
3. **Spatial coherence**
 1. zones are “linearly separated” on the map
 2. **positive spatial autocorrelation**: neighboring nodes have similar prices

Results

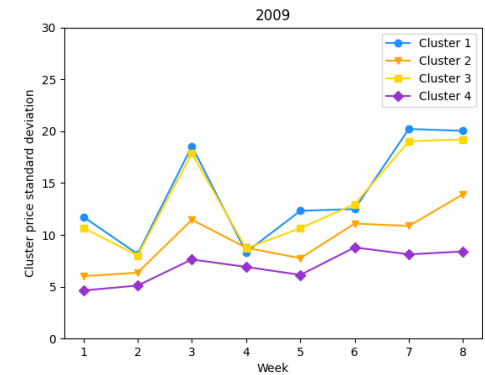
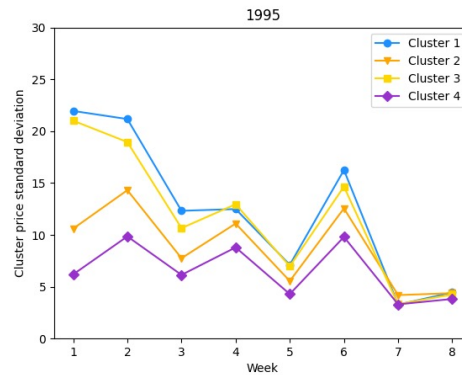
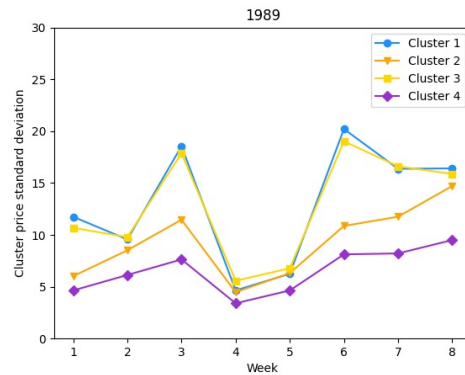
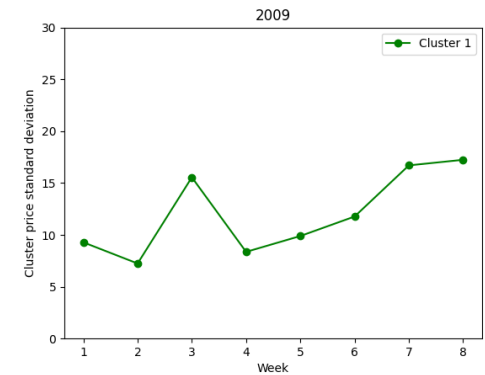
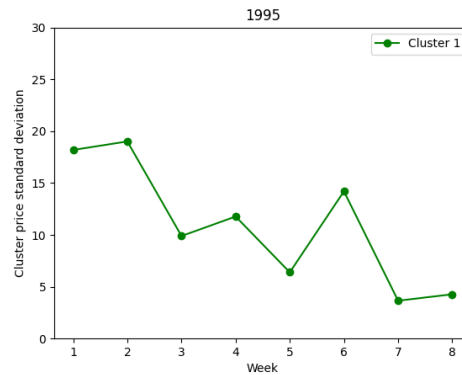
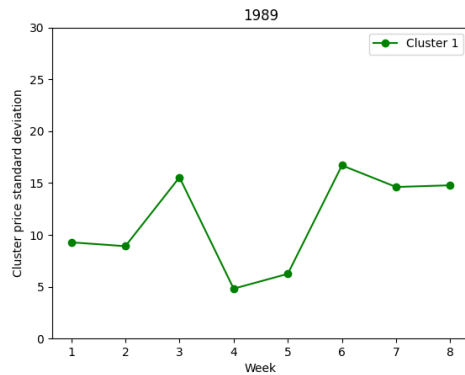
Intra-cluster similarity: Price std – DE2 (k-means)



Intra-cluster similarity: Price std – DE3 (spectral)



Intra-cluster similarity: Price std – DE4 (spectral)



Intra-cluster similarity: Average price std

	1989	1995	2009	% decrease
Single price zone	14.54	14.32	11.48	
DE2 (k-means)	13.09	12.93	10.85	8.55
DE2 (spectral)	13.98	13.80	11.22	3.27
DE3 (spectral)	13.86	13.65	11.26	3.86
DE4 (spectral)	13.14	13.02	10.87	8.18

Inter-cluster dissimilarity: Average cluster prices (I)

Single price zone:

		W1	W2	W3	W4	W5	W6	W7	W8	Δ^{year}
1989	Cluster 1	50.49	46.72	41.99	49.88	48.52	39.52	40.22	45.50	-
1995	Cluster 1	39.87	28.09	45.89	46.11	48.61	38.35	51.76	54.50	-
2009	Cluster 1	50.49	51.94	41.99	47.66	45.89	46.11	39.52	40.91	-

DE2 (k-means):

$$\Delta_{\mu}^{W_1} = 3.46$$

		W1	W2	W3	W4	W5	W6	W7	W8	$\Delta_{\mu}^{\text{year}}$
1989	Cluster 1	49.09	45.71	39.18	49.32	47.94	35.84	37.44	44.38	4.3
	Cluster 2	52.55	48.21	46.10	50.69	49.39	44.91	44.28	47.15	
1995	Cluster 1	35.02	21.21	44.56	45.13	47.65	35.63	51.22	54.14	5.75
	Cluster 2	46.98	38.19	47.84	47.55	50.03	42.35	52.56	55.04	
2009	Cluster 1	49.09	51.21	39.18	47.11	44.56	45.13	35.84	37.25	4.67
	Cluster 2	52.55	53.01	46.10	48.46	47.84	47.55	44.91	46.28	

Inter-cluster dissimilarity: Average cluster prices (II)

Single price zone:

		W1	W2	W3	W4	W5	W6	W7	W8	Δ_{year}
1989	Cluster 1	50.49	46.72	41.99	49.88	48.52	39.52	40.22	45.50	-
1995	Cluster 1	39.87	28.09	45.89	46.11	48.61	38.35	51.76	54.50	-
2009	Cluster 1	50.49	51.94	41.99	47.66	45.89	46.11	39.52	40.91	-

DE4 (spectral):

		W1	W2	W3	W4	W5	W6	W7	W8	$\Delta_{\mu}^{\text{year}}$
1989	Cluster 1	49.03	46.0	39.51	49.91	48.52	36.68	38.28	44.48	3.14
	Cluster 2	52.30	47.85	45.41	50.94	49.77	44.74	43.06	46.37	
	Cluster 3	49.00	45.41	38.73	48.84	47.44	34.78	36.55	44.25	
	Cluster 4	52.62	48.50	46.44	50.37	48.91	44.81	45.46	47.86	
1995	Cluster 1	35.24	21.24	44.78	45.75	48.32	36.11	51.63	54.13	4.19
	Cluster 2	45.35	35.98	47.97	47.80	50.53	41.13	53.33	55.07	
	Cluster 3	34.37	20.74	44.23	44.60	47.05	34.99	50.87	54.11	
	Cluster 4	48.32	39.51	47.58	47.13	49.35	43.53	51.58	54.94	
2009	Cluster 1	49.03	51.29	39.51	47.72	44.78	45.75	36.68	37.71	3.34
	Cluster 2	52.30	53.05	45.41	49.11	47.97	47.80	44.74	44.95	
	Cluster 3	49.00	51.07	38.73	46.6	44.23	44.60	34.78	36.69	
	Cluster 4	52.62	52.87	46.44	47.67	47.58	47.13	44.81	47.18	

Temporal stability

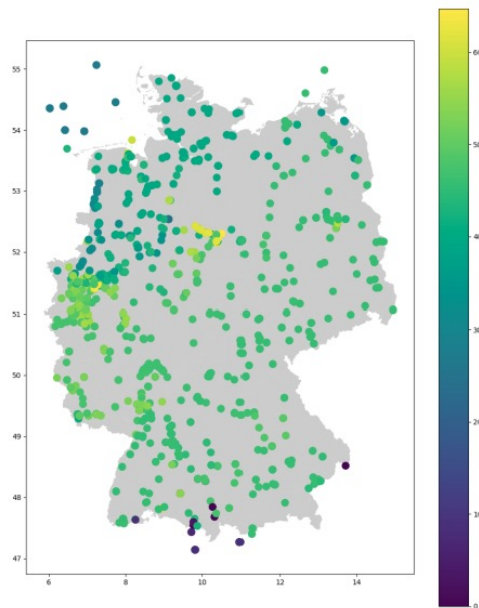
Rand indices*:

		1989	1995	2009
K-Means	DE2 (k-means)	0.5	0.5	0.51
	DE2 (spectral)	0.5	0.5	0.5
	DE3 (spectral)	0.51	0.53	0.52
	DE4 (spectral)	0.6	0.5	0.56
Spectral	DE2 (k-means)	0.51	0.51	0.51
	DE2 (spectral)	0.5	0.5	0.5
	DE3 (spectral)	0.52	0.44	0.52
	DE4 (spectral)	0.55	0.45	0.57

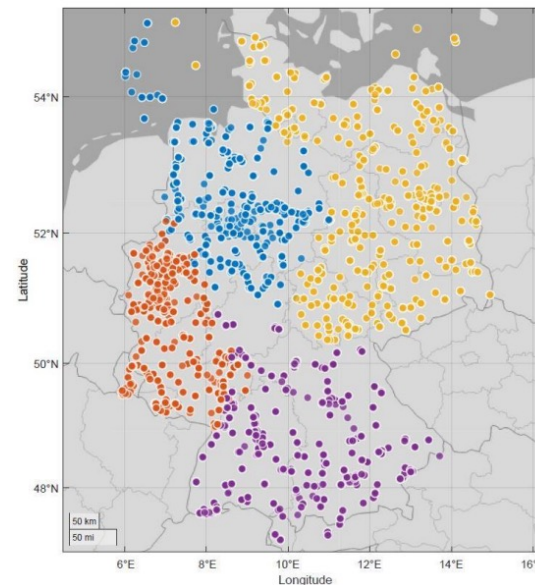
* $RI \in [0, 1]$: 1 = perfect match

Spatial coherence

- ✓ Clusters are linearly separated on the map
- ✓ Positive spatial autocorrelation, but no well-defined zones w.r.t. prices



Average nodal prices on 10.03.2009



Intermediate summary – ACER configurations

1. Clustering based on **latitude & longitude** → clusters similar to ACER's proposals.
2. The **average decrease** in **price standard deviation** inside clusters is **< 9%**.
 - on average: 6%.
3. The **difference** in the **cluster average prices** is **low**.
 - < 6 EUR/MWh
 - on average: 4.16 EUR/MWh.
4. The configurations are **not stable over time**.
5. The configurations are **spatially coherent**.

Other configurations?

Based (only) on prices, can we find other configurations that are **more stable**?

Other configurations?

Based (only) on prices, can we find other configurations that are **more stable**? **NO**

- **higher intra-cluster similarity, but...**

Year	k	Alg.	Loc.	σ	Δ_μ	$\Delta_{\min}$	Δ_{avg}	DB	μ_1	μ_2	μ_3	μ_4
1989	2	K-Means		13.4	5.91	0.0	5.91	0.63	47.28	41.37	-	-
1995	2	K-Means		11.16	4.66	0.0	4.97	0.76	48.47	43.81	-	-
2009	2	K-Means		10.8	3.68	0.0	3.68	0.71	45.29	48.97	-	-
1989	2	Spectral		13.39	3.57	0.0	4.3	1.44	43.61	47.18	-	-
1995	2	Spectral		15.16	2.87	0.0	3.57	1.89	44.74	41.87	-	-
2009	2	Spectral		11.61	0.31	0.0	2.19	13.76	46.68	46.37	-	-
1989	2	K-Means	x	13.0	5.91	0.0	5.93	0.62	41.64	47.55	-	-
1995	2	K-Means	x	13.03	5.9	0.0	5.93	0.56	47.88	41.98	-	-
2009	2	K-Means	x	9.97	2.44	0.0	2.91	0.84	46.36	48.8	-	-
1989	2	Spectral	x	15.46	1.29	0.0	3.97	5.72	44.05	42.76	-	-
1995	2	Spectral	x	14.2	3.0	0.0	4.41	2.26	44.11	47.11	-	-
2009	2	Spectral	x	11.68	0.39	0.0	2.25	9.3	46.64	46.25	-	-

Other configurations?

Based (only) on prices, can we find other configurations that are **more stable**? **NO**

– low inter-cluster dissimilarity

Nearest neighbor cluster distance

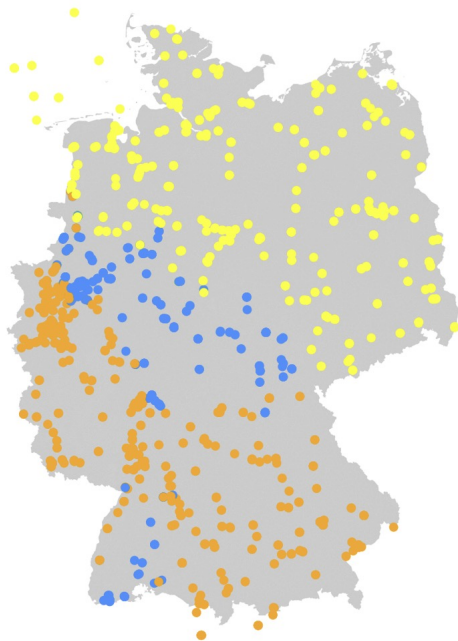
Year	k	Alg.	Loc.	σ	Δ_{μ}	$\Delta_{\min}$	Δ_{avg}	DB	μ_1	μ_2	μ_3	μ_4
1989	2	K-Means		13.4	5.91	0.0	5.91	0.63	47.28	41.37	-	-
1995	2	K-Means		11.16	4.66	0.0	4.97	0.76	48.47	43.81	-	-
2009	2	K-Means		10.8	3.68	0.0	3.68	0.71	45.29	48.97	-	-
1989	2	Spectral		13.39	3.57	0.0	4.3	1.44	43.61	47.18	-	-
1995	2	Spectral		15.16	2.87	0.0	3.57	1.89	44.74	41.87	-	-
2009	2	Spectral		11.61	0.31	0.0	2.19	13.76	46.68	46.37	-	-
1989	2	K-Means	x	13.0	5.91	0.0	5.93	0.62	41.64	47.55	-	-
1995	2	K-Means	x	13.03	5.9	0.0	5.93	0.56	47.88	41.98	-	-
2009	2	K-Means	x	9.97	2.44	0.0	2.91	0.84	46.36	48.8	-	-
1989	2	Spectral	x	15.46	1.29	0.0	3.97	5.72	44.05	42.76	-	-
1995	2	Spectral	x	14.2	3.0	0.0	4.41	2.26	44.11	47.11	-	-
2009	2	Spectral	x	11.68	0.39	0.0	2.25	9.3	46.64	46.25	-	-

Average distance between
cluster mean prices

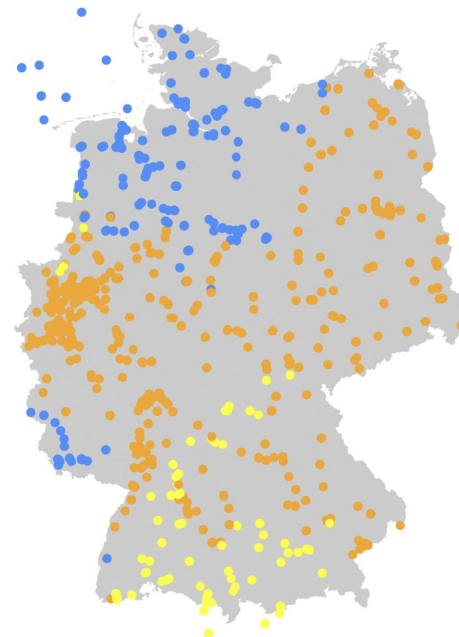
Other configurations?

Based (only) on prices, can we find other configurations that are **more stable**? **NO**

- **low spatial coherence**



K-Means, 1989

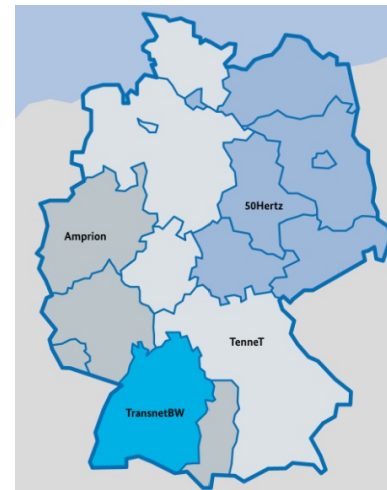
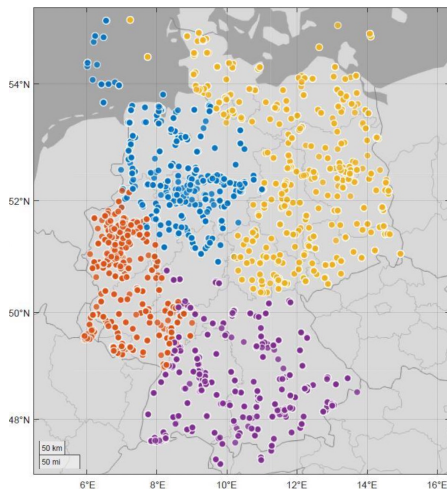


Spectral Clustering, 1995

Summary

- Finding stable price zones based only on prices seems to be a moving target.
- Multiple factors should be considered:
 - Geographical coherence, federal structure
 - TSO's control areas
 - Physical infrastructure, cross-zonal trade
 - Generation & demand patterns, etc.

Nodal market design?



Source: <https://www.smard.de/page/en/wiki-article/5884/205528>

Thank you!